

- # A push or a pull is a force in science.
- # ~~Reset~~ When force is applied on a object, it may:
 - Object may move from rest
 - If object is moving, its speed might change
 - Change in the direction of motion
 - Bring a change to its shape
- # It happens by the interaction of an obj. with another obj.

SI unit of force is Newton (N)

When there is physical contact b/w the obj's, then this force is 'Contact force'

Eg. of Contact Force - Friction

Muscular force

The force generated by our muscles through contraction & relaxation while doing any activity.

Eg: ~~Heart~~ Heart, chewing

The force which acts in the direction opp. to the direction of moving obj. It reduces its speed & stops it. It arises due to the irregularities b/w 2 surfaces

All liquids & air also exert friction of obj. moving through them. That's why planes, ships, boat & high-speed train all have special design.

Non-contact force

The force which can be felt even without contact

Magnetic force

The force exerted by an magnet on another magnet or magnetic substance
Eg: Repulsion b/w N & N.

Electrostatic Force

The force exerted on a charged body by another body or an uncharged body.

Gravity

The force by which Earth pulls everything towards itself. When an object is falling, it undergoes vertical motion as it moves in a vertical path down.

Electrostatic Force : In deep

- When 2 objects are rubbed, electrical charges built-up, as they can't move on their own, they are known as static charges.
- The obj. which has static charges is called charged obj.
- Similarly charged objects repel, but if one obj. is uncharged & other is charged, it attracts.
- It generates electric current too.

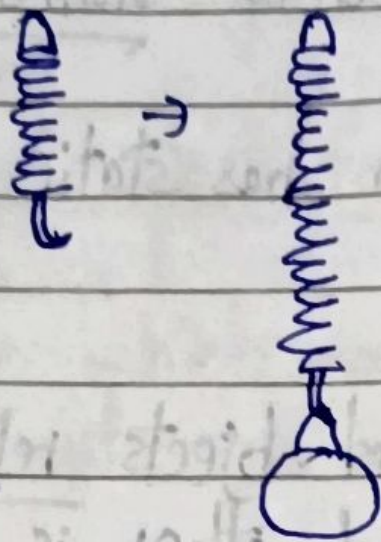
The force by which the Earth pulls an obj. towards itself is called weight.
It shows how ~~gr~~ strongly gravity pulls it.

Formula: $W = \text{mass} \times \text{gravity}$

It is also measured in N.

A spring balance is a simple device used ~~mean~~ measure the weight of any obj.
It has a spring at one end, & a hook at other end. When we hang any obj. to the hook, the spring stretches. The amount of

stretching give the weight of the obj. as the values are already marked. It shows weight in Newtons & values of mass in grams.



How to measure the smallest value that can be measured on a spring balance

→ Step 1: count the no. of ~~values~~ marks b/w 1 N & 2 N.

Step 2: Divide that no. by 1 N.

$$\text{Eg: } \frac{1 \text{ N}}{5} = \boxed{0.2 \text{ N}}$$

Weight

It is the force by which gravity attracts any object. Changes depending upon the place.

Mass

Amount of matter in an object. It is measured in Grams / Kgs. It never change.

Usually, we consider weight & mass in everyday life, but scientifically it is wrong.

Upthrust or Buoyant force is a force applied by every liquid. When you place an object in a liquid, they are pulled down by gravity & this force pushes them up.

It is one of the facts that decide that "Will an object float or not?" as if upthrust is greater than gravity, it will float.

Archimedes, was a greek scientist who discovered the fact that when an obj is immersed in water, it experience an upward force which is equal to the weight ~~amount~~ of water displaced by the obj. If, the amount of displaced water is equal to weight of the obj, the obj. 'll float.

Pumice is a very light & porous filled with air pockets in it. It is made during volcanic eruptions.